

ASSESSMENT TOOL – 3 CO1
TRANSFORMER WORKSHEET

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Course Code : CSA6502

Course Name : Generative AI For Large Scale Models

Question 1: Tokenization

(a) Split "DISAGREEMENT" into 3 reasonable sub-word pieces :

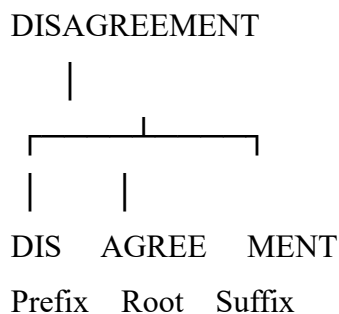
DIS + AGREE + MENT

Justification

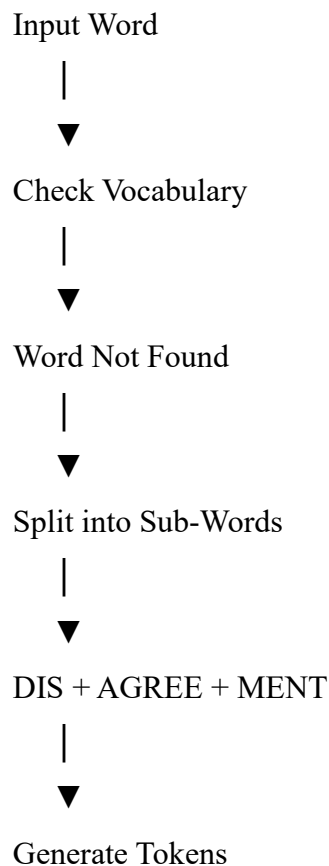
- DIS → Prefix meaning "not" or "opposite."
- AGREE → Root word.
- MENT → Suffix that converts a verb into a noun.

Sub-word tokenizers (like BPE or WordPiece) often split uncommon words into meaningful pieces that already exist in their vocabulary.

Structure Representation :



Flowchart :



(b) Why is sub-word tokenization useful?

Answer

- (c) Sub-word tokenization helps the model understand new or rare words by breaking them into familiar parts. Even if the model has never seen "DISAGREEMENT", it already knows the meanings of DIS, AGREE, and MENT, allowing it to understand the whole word.

Theory

- (d) Instead of storing millions of complete words, transformers store smaller word pieces. This reduces vocabulary size and helps handle unknown words.

Conclusion

- (e) Sub-word tokenization improves vocabulary coverage, reduces unknown words, and helps the model understand new words effectively.

Question 2: Embeddings

(a) Which pair has higher cosine similarity?

Pairs:

- Dog – Puppy
- Dog – Truck

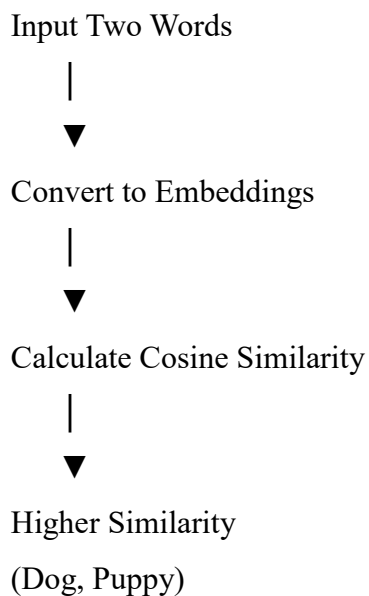
Answer

Dog and Puppy

Explanation

- Dog and Puppy are semantically related. Their embeddings are close together in vector space.
- Dog and Truck belong to different categories, so their embeddings are far apart.

Distance :



(b) What is an embedding? How is it different from a one-hot vector?

Answer

- An embedding is a dense numerical vector that represents the meaning of a word. Similar words have similar embeddings.

- A one-hot vector contains only one value equal to 1 and all other values equal to 0, so it cannot represent similarity between words.

Theory

- One-Hot Vector
- Dog

[0 0 1 0 0]

Contains learned semantic information.

Comparison Table :

Feature	One-Hot Vector	Embedding
Size	Very Large	Small
Meaning	None	Semantic
Similarity	No	Yes
Learning	Fixed	Learned

Conclusion

Embeddings capture semantic relationships, whereas one-hot vectors only identify words.

Question 3: Self-Attention Worked Example :

Sentence:

"The city council refused the protesters a permit because they feared violence."

(a) "They" refers to?

Answer

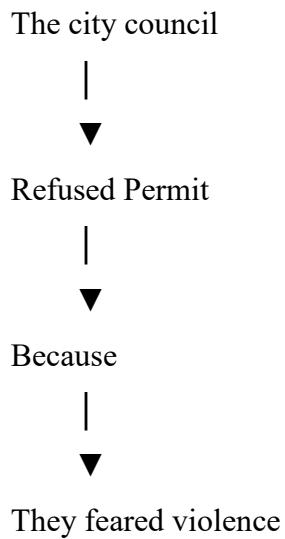
City council

Explanation

The sentence suggests that the city council refused the permit because they feared violence.

Structure:

Sentence



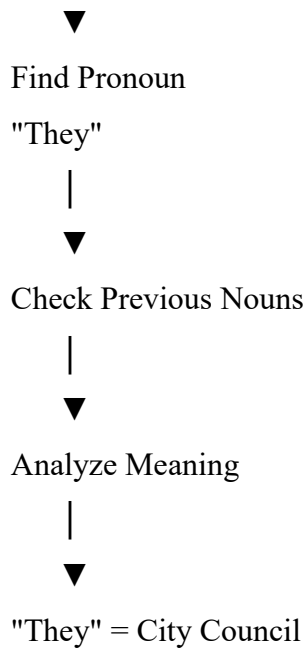
"They"



Flowchart :

Read Sentence





(b) Why is self-attention needed?

Answer

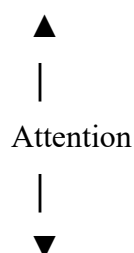
Self-attention allows the model to connect the word "they" with the correct earlier words in the sentence based on meaning, not just position. It examines relationships between all words to determine the correct reference.

Theory

- Traditional models read words one after another.
- Transformers compare every word with every other word.
- This allows them to resolve pronouns correctly.

Self-Attention Representation :

The city council refused the protesters



They feared violence

Conclusion

Self-attention helps transformers understand context and identify the correct meaning of pronouns.

Question 4: Query, Key, Value (5 Marks)

Answer

Query (Q)

The Query represents what the current word is searching for.

Key (K)

The Key represents what information each word contains.

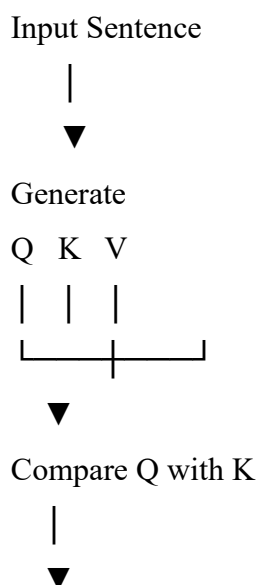
Value (V)

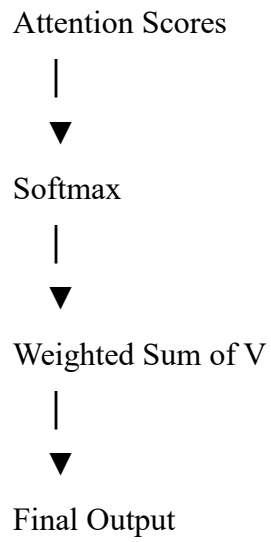
The Value contains the actual information that will be passed to the output.

How Attention Works

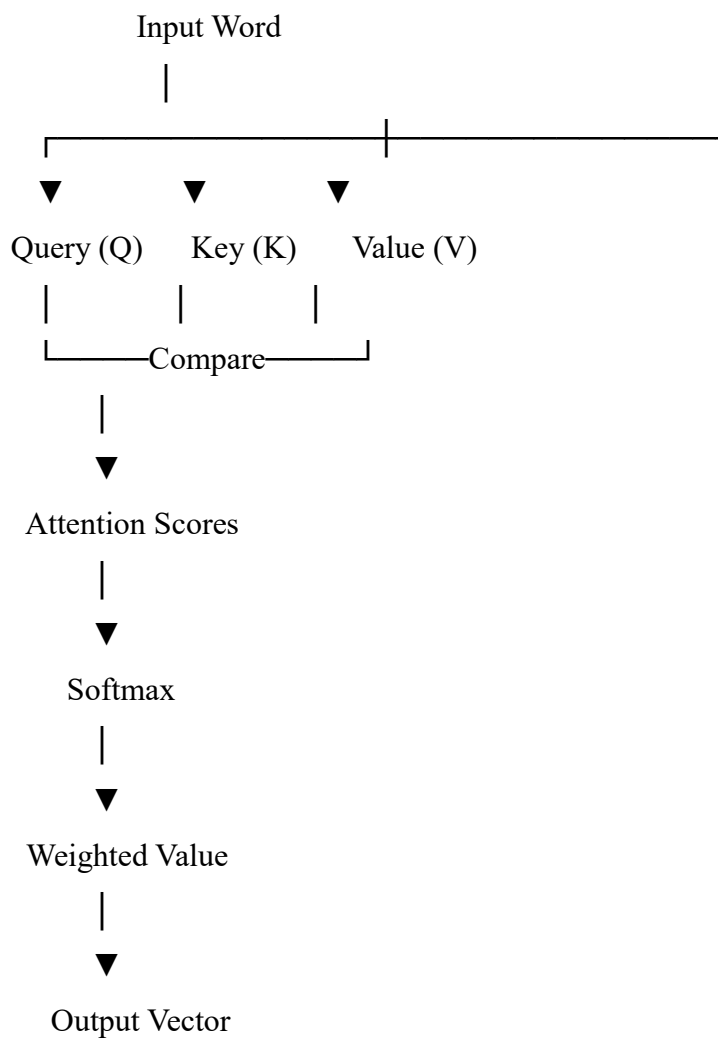
1. Create Query, Key and Value vectors.
2. Compare Query with all Keys.
3. Compute similarity scores.
4. Apply Softmax.
5. Multiply attention weights with Values.
6. Produce final output.

Flowchart :





Structure Diagram :



Mathematical Representation :

$$\text{Attention}(Q,K,V)$$

$$\frac{Q \times K^T}{\sqrt{d_k}} \times V$$

Where:

- Q = Query Matrix
- K = Key Matrix
- V = Value Matrix
- $\sqrt{d_k}$ = Scaling factor
- Theory
- Query asks: *What information do I need?*
- Key answers: *What information do I have?*
- Value provides: *The information that will be used.*

The transformer compares Query with Keys to decide how much attention each word should receive, then combines the corresponding Values.

Conclusion

The Query-Key-Value mechanism enables transformers to focus on the most relevant words in a sentence, improving understanding of context, relationships, and meaning. It is the core idea behind the self-attention mechanism and contributes significantly to the high performance of transformer models in natural language processing.